CCEE Exam Prep: Git & GitHub Cheat Sheet

1 1. The Core Distinction: Git vs. GitHub

Crucial for Viva/Theory: Do not confuse the tool with the service.

Feature	Git	GitHub
Definition	Version Control System (VCS) Tool	Hosting Service for Git Repositories
Location	Installed locally on your computer	Website/Cloud Server
Connectivity	Works 100% Offline	Requires Internet connection
Primary Role	Tracks changes, manages history	Hosting, Collaboration, Backup

2 2. The Git Lifecycle (The "Three Stages")

You must know where your file resides at any given moment.

1. **Working Directory:** Where you edit files (Untracked/Modified).
2. **Staging Area (Index):** Where you prepare files for saving (`git add`).
3. **Local Repository:** Where files are permanently saved (`git commit`).

> **The Golden Flow:**
> Edit File → `git add` → Staging → `git commit` → Local Repo → `git push` → Remote

3 3. Essential Command Reference

3.1 Setup & Initialization

- `git config --global user.name "Name"` – Sets your username.
- `git init` – Initializes a new repository (creates hidden `.git` folder).
- `git clone <url>` – Copies an existing repo from GitHub to your machine.

3.2 The Daily "Save" Cycle

- `git status` – **Most Important.** Checks what is modified or staged.
- `git add <file>` – Moves a specific file to Staging.
- `git add .` – Moves **all** changes to Staging.
- `git commit -m "message"` – Saves staged files to Local Repo.
- `git log` – Shows the history of commits.

3.3 Syncing with GitHub (Remote)

- `git remote add origin <url>` – Links local repo to GitHub.
- `git push -u origin main` – Uploads local commits to GitHub (first time).
- `git pull` – Downloads changes AND merges them (Fetch + Merge).
- `git fetch` – Downloads changes but does NOT merge (Safer).

3.4 Branching (Parallel Development)

- `git branch` – Lists all branches.
- `git branch <name>` – Creates a new branch.
- `git checkout <name>` – Switches to a branch.
- `git merge <name>` – Joins the specified branch into current branch.

4 4. Key Exam Concepts

- **.gitignore:** A text file listing file patterns Git should ignore (e.g., `*.log`, `node_modules/`).
- **Merge Conflict:** When Git cannot automatically join differences (e.g., same line changed differently). Requires manual fix.
- **HEAD:** A pointer to the latest commit of the currently checked-out branch.